

Solid State Physics

PHYS 412

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Electrons in Periodic Potential

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- Previously we have looked at Fermi Gas, Sommerfeld Model (Drude picture in Fermi Gas) but we didn't consider any lattice system. Tight Binding showed Bonds $\rightarrow$ energy bands, and there we get the notion of "periodicity" with particular band gap features showing at the spatial frequencies π/a . When we consider lattice, the Reciprocal Lattice is an important idea. For instance Bragg Reflection of waves happens when change in a wavevector is a reciprocal lattice vector.
- The lattice problem can be treated as a perturbation to the general problem. Solution to unperturbed and perturbed system (where $H = H_0 + \lambda V_1$, unperturbed and perturbed are H_0 and λV_1 respectively).

$$H_0\psi_0(x)^{(n)} = E_0^{(n)}\psi_0(x)^{(n)}$$

$$H\psi(x) = E(x)$$

We have solved the problem of finding the energy corrections (or perturbations) in Perturbation Theory for Quantum Mechanics (PHYS 312).

$$E_1 = \langle \psi_0 | V | \psi_0 \rangle$$

$$|\psi_1\rangle = \sum_{m \neq n} \frac{\langle \psi^m | V | \psi^n \rangle}{E^{(n)} - E^{(m)}} |\psi^{(m)}\rangle$$

- Fermi gas is the unperturbed starting point. Here the Hamiltonian looks like the Free Particle.

$$H_0 = \frac{p^2}{2m} \quad \varepsilon_0(k) = \frac{\hbar^2 k^2}{2m}$$

The states are $|\mathbf{k}\rangle = \frac{1}{\sqrt{L^3}} e^{i\mathbf{k}\cdot\mathbf{r}}$. Now invoke a potential $V(r)$ that has the property $V(\mathbf{r} + \mathbf{R})$. The matrix element of the perturbation is given by

$$\langle \mathbf{k}' | V | \mathbf{k} \rangle = \int d\mathbf{r} \frac{e^{-i\mathbf{k}'\cdot\mathbf{r}}}{\sqrt{L^3}} V(\mathbf{r}) \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{L^3}} = \frac{1}{L^3} \int d\mathbf{r} e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} V(\mathbf{r}) = V_{\mathbf{k}'-\mathbf{k}}$$

This given integral is zero unless $\mathbf{k} - \mathbf{k}' = \mathbf{G}$.

- Any function with real-space periodicity of the lattice can be built out of an appropriate combination of reciprocal lattice vectors

$$f(\mathbf{r}) = f(\mathbf{r} + \mathbf{R}) = \sum_{\mathbf{G}} f_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{r}}$$

The non-vanishing Fourier components of the lattice potentials are the reciprocal lattice vectors.

- General Perturbation Theory result

$$\epsilon(\mathbf{k}) = e_0(\mathbf{k}) + \langle \mathbf{k} | V | \mathbf{k} \rangle + \sum_{\mathbf{k} \neq \mathbf{k}'} \frac{|\langle \mathbf{k}' | V | \mathbf{k} \rangle|^2}{\epsilon_0(\mathbf{k}) - \epsilon_0(\mathbf{k}')}$$

Now we want to take the sum like $\mathbf{k}' = \mathbf{k} + \mathbf{G}$ for non-zero $\mathbf{G}$ but this blows up when we reach near the edge of the lattice ($\epsilon_0(\mathbf{k}) = \epsilon_0(\mathbf{k}')$). Which is a motivation to use Degenerate Perturbation Theory at the zone boundary.

- We define the matrix elements $\langle \mathbf{k}' | V | \mathbf{k}' \rangle, \langle \mathbf{k} | V | \mathbf{k} \rangle, \langle \mathbf{k}' | V | \mathbf{k} \rangle, \langle \mathbf{k} | V | \mathbf{k}' \rangle$ such that

$$\begin{aligned} \langle \mathbf{k} | H | \mathbf{k} \rangle &= \epsilon_0(\mathbf{k}) \\ \langle \mathbf{k}' | H | \mathbf{k}' \rangle &= \epsilon_0(\mathbf{k}') = \epsilon_0(\mathbf{k} + \mathbf{G}) \\ \langle \mathbf{k} | H | \mathbf{k}' \rangle &= V_{\mathbf{k}-\mathbf{k}'} = V_G^* \\ \langle \mathbf{k}' | H | \mathbf{k} \rangle &= V_G \end{aligned}$$

- The bigger the Fourier Gap the bigger the energy band gap (?)
- In one 1D $V(x) = V_0 \cos(2\pi x/a)$, at zone boundary we will get
- Perturbed wavefunctions always look like Bloch Waves.
- Blochs Theorem
- Bloch States are not momentum eigenstates. $\hbar\mathbf{k}$ is the crystal momentum. This can be observed by operating the usual momentum operator $\mathcal{P} = -i\hbar\nabla$